

Identifying the Distribution of Baseline Noise in Mass Spectrometer Measurements

Client: Thermochron Systems - Adam Goldsmith

Course: STAT 192 Statistics Capstone Project

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1. Summary

This project analyzes baseline noise in Prisma Pro mass spectrometer measurements so future experimental signals can be corrected more accurately. Baseline Characterization Analysis (BCA) files provide direct measurements of instrument noise at six dwell times, while real experiments contain signal channels plus a live baseline channel.

The cleaned BCA dataset contains 492,000 observations, with exactly 82,000 observations at each dwell time: 0.128, 0.256, 0.512, 1.024, 2.048, and 4.096 seconds. Exploratory analysis shows that baseline noise is positive-valued, right-skewed, heavy-tailed, and strongly dependent on dwell time. As dwell time increases, the center and spread of the noise distribution decrease.

Because the noise distribution changes by dwell time, a single pooled baseline correction is not appropriate. The recommended modeling strategy is a reparameterized log-skew-t distribution in which location, scale, and tail-heaviness are modeled as functions of dwell time. A single shared skewness parameter is estimated across dwell times. This structure captures the observed dwell-time dependence while avoiding a separate fully unconstrained distribution fit for every dwell-time group.

The fitted model supports dwell-time-specific correction as the main conclusion. The live baseline channel should be retained as useful diagnostic information, but weak cross-dwell correlations mean it should not automatically be used as a direct cycle-level correction for every other dwell time.

2. Introduction

2.1 Background

Geochronology by gas mass spectrometry estimates rock ages from the relative abundances of gaseous isotopes released from a sample. In this setting, the mass spectrometer records an electrical current, measured in amperes (A), that corresponds to the intensity of ions detected at a specified mass-to-charge ratio. During measurement, the sample is introduced into the mass spectrometer, the mass filter selects ions with a targeted mass-to-charge ratio, and the selected ions strike a detector. The detector converts these ion impacts into an electrical current, which is recorded as the measurement intensity. In the data considered for this project, typical recorded currents are very small, ranging approximately from $10^{-16} A$ to $10^{-13} A$.

However, the recorded current is not necessarily equal to the true sample signal. Even when no sample ions are present, the instrument may record a nonzero electrical current. This background contribution is referred to as baseline noise. Baseline noise can arise from several sources, including electronic noise in the detector, stray ions reaching the detector, and environmental or instrument-level fluctuations. Because this noise is present during measurement, accurate estimation of the true sample signal requires separating the observed current into a signal component and a noise component.

For an individual observation i recorded at dwell time d , the measurement can be represented as

$$Y_{raw,i}(d) = Y_{signal,i}(d) + N_i(d) \quad (2.1)$$

where $Y_{raw,i}(d)$ is the recorded electrical current, $Y_{signal,i}(d)$ is the true sample-generated current, and $N_i(d)$ is the baseline-noise contribution. The notation d is included because the noise distribution may depend on dwell time, meaning that baseline noise measured at one dwell time may not be representative of baseline noise at another dwell time.

A common baseline-correction approach subtracts a live baseline reading from the raw measured signal. In simplified form, this correction can be written as

$$\hat{Y}_{signal,i}(d) = Y_{raw,i}(d) - \hat{N}_i(d) \quad (2.2)$$

where $\hat{N}_i(d)$ is an estimate of the baseline-noise contribution. This approach is reasonable only if the estimated baseline noise is measured under conditions that are comparable to the target signal measurement. If the baseline estimate is too large, or if it is measured at a dwell time that does not match the target signal channel, the corrected value may become negative:

$$Y_{raw,i}(d) - \hat{N}_i(d) < 0 \quad (2.3)$$

Such negative corrected currents are physically implausible when the quantity of interest is the true ion current. This motivates the need to characterize the baseline-noise distribution directly, especially its dependence on dwell time, before proposing or applying a correction method. A reliable correction method should estimate and remove the machine-level background noise while preserving physically meaningful, nonnegative estimates of the true sample signal.

2.2 Baseline Characterization Analysis

A Baseline Characterization Analysis (BCA) run is designed to measure instrument background noise directly. In a BCA run, the mass spectrometer is configured to monitor a mass channel

where no real gas species is expected to be present. Under this configuration, the true sample-generated signal is assumed to be zero:

$$Y_{signal,i}(d) = 0 \quad (2.4)$$

Substituting Equation (2.4) into the general measurement model from Equation (2.1) gives

$$Y_{raw,i}(d) = N_i(d) \quad (2.5)$$

A BCA file provides observations of the baseline-noise distribution at each dwell time supported by the instrument configuration. Because the dwell-time channels are recorded in parallel within the same run, these measurements are collected under the same instrument and environmental conditions. This structure makes BCA data especially useful for evaluating whether the noise distribution changes systematically with dwell time. The collection of BCA files across instruments and laboratories forms the empirical basis for the distributional analyses in this report.

In standard experimental measurements, the observed current combines true sample signal and baseline noise, as shown in Equation (2.1). In contrast, a BCA run is intended to remove the sample-signal component by design, allowing the observed current to be treated as an approximate realization of $N_i(d)$. This makes the BCA data suitable for estimating dwell-time-specific features of the baseline-noise distribution, including location, spread, skewness, tail behavior, and candidate parametric or non-parametric model forms.

Each BCA file records parallel noise streams, with one stream corresponding to each available dwell time. For the Prisma Pro instrument, the supported dwell times are

$$d \in \{0.128, 0.256, 0.512, 1.024, 2.048, 4.096\} \text{ seconds.} \quad (2.6)$$

For the Prisma Plus instrument, the supported dwell times are

$$d \in \{0.1, 0.2, 0.5, 1, 2, 5\} \text{ seconds.} \quad (2.7)$$

The present analysis focuses on the Prisma Pro BCA data. Therefore, unless otherwise stated, references to dwell time in the remainder of this report refer to the Prisma Pro dwell-time grid in Equation (2.6).

2.3 Standard Experimental Analysis

A standard experimental analysis differs from a Baseline Characterization Analysis because the instrument is configured to measure ions from real sample gases. In a standard run, each measurement channel is assigned to a targeted mass channel corresponding to an isotope or gas species of analytical interest. The recorded current in these target channels therefore contains

both the true sample-generated signal and the background noise contribution described in Equation (2.1):

$$Y_{raw,i}(d_c) = Y_{signal,i}(d_c) + N_i(d_c) \quad (2.8)$$

where c indexes the measurement channel and d_c denotes the dwell time assigned to channel c .

Unlike a BCA run, the signal component $Y_{signal,i}(d_c)$ is not assumed to be zero for target isotope channels.

For each target channel, the analyst assigns a dwell time based on the expected abundance of the isotope, the desired measurement precision, and the analytical priority of that channel. Longer dwell times generally allow the instrument to integrate the signal over a longer period, while shorter dwell times allow faster cycling across channels. As a result, different isotope channels within the same experimental run may be measured at different dwell times.

In the standard experimental files considered for this project, the fourth channel, corresponding to the fourth intensity column in the data, is configured as a live baseline channel. This channel is fixed at a mass setting of 5.5 atomic mass units (amu), where no real gas species is expected to be present, under the same logic used for BCA measurements. The live baseline channel is measured at a dwell time of 0.512 seconds.

2.4 Dataset

The dataset used in this report is drawn from dedicated Baseline Characterization Analysis (BCA) runs collected on three Prisma Pro quadrupole mass spectrometers. The cleaned, pooled dataset contains

$$n = 492,000 \quad (2.9)$$

baseline-noise observations.

The data are organized by laboratory source and experimental batch. Four source directories are included in the project archive: Glasgow, Salzburg, Wuhan, and Nineamu. However, Nineamu is treated as a separate experimental batch rather than a distinct laboratory, because it was completed at Salzburg, it represents a distinct experimental batch with a different target amu configuration. Therefore, the dataset represents BCA measurements from three laboratory locations, with four source directories used for file organization and batch tracking.

Since BCA channels are configured to measure the same baseline mass setting, the channel identities are not distinguished by target isotope or target mass. Instead, each intensity column is mapped to its corresponding dwell time.

The primary variables used in the analysis are:

$$time_sec, d, Y_{raw,i}(d) \quad (2.10)$$

Here, $time_sec$ records the timestamp within a BCA run, d denotes the dwell time associated with the measurement channel, and $Y_{raw,i}(d)$ denotes the recorded current for observation i at dwell time d .

Files are included from the Glasgow, Salzburg, Wuhan, and Nineamu source directories. Nineamu is retained as a separate batch label because it is a distinct experimental batch. Preserving this label allows the analysis to evaluate whether the baseline-noise distribution varies by laboratory source, experimental batch, or target amu setting.

2.5 Data Preprocessing

The preprocessing pipeline converts semi-structured BCA instrument files into analysis-ready long-format and wide-format datasets. The main steps include identifying valid source files, parsing timestamp and intensity columns, classifying files by experiment type, correcting known acquisition artifacts, handling timestamp resets, and exporting cleaned data objects for downstream analysis. The following subsections describe each step in detail.

2.5.1 File Parsing

Raw data files are stored across multiple laboratory directories and follow a semi-structured text format. The raw files are drawn from the project's `/BCA` directory. Only files matching the following regular expression pattern are selected.

$$He[0-9]^+ .txt \quad (2.11)$$

Each candidate file is read as a tabular text file. The first row contains column labels and instrument metadata that must be interpreted before the file can be converted into an analysis-ready data frame. The parser extracts the timestamp column, denoted as $time_sec$, and the intensity columns corresponding to the instrument's measurement channels. Each intensity column contains the recorded electrical current for one channel at each timestamp.

Because BCA files are configured so that all measurement channels are assigned to the same baseline mass setting, the intensity columns are not treated as distinct target-isotope channels. Instead, each intensity column is relabeled according to its assigned dwell time d . This relabeling is necessary because the primary objective of the analysis is to evaluate the dwell-time-specific noise process $N_i(d)$, rather than isotope-specific signal behavior.

Files that do not conform to the expected BCA structure are excluded from further analysis. Exclusion criteria include missing or malformed headers, irregular column counts, incompatible delimiters, and file structures that cannot be reliably mapped to timestamp and intensity columns.

After ingestion, each file is classified based on its structure:

- **BCA:** all six dwell-time channels are tuned to the same baseline mass; six intensity columns per row.
- **PBCA:** the same configuration with five intensity columns. PBCAs represent a separate experimental configuration that did not yield usable data and are excluded from this analysis.
- **Experimental:** dwell-time channels are tuned to different masses; the column header lists distinct mass labels

This restriction ensures that the modeled observations correspond to uncontaminated baseline-noise measurements under the BCA assumption in Equation (2.5). The full raw-file structure and variable definitions are provided in the data dictionary in Appendix C.

2.5.2 Automatic Baseline Reversal

Some BCA files were affected by an instrument-level automatic baseline correction that had not been disabled during data collection. When this setting is active, the instrument subtracts a designated baseline channel from the recorded intensity channels during acquisition. For a BCA run, this behavior is undesirable because the purpose of the run is to observe the raw baseline-noise process directly. If the automatic correction is applied before export, the resulting file no longer contains direct observations of $N_i(d)$.

Let $Y_{raw,i}(d_c)$ denote the uncorrected BCA intensity for observation i in channel c , and let B_i denote the baseline channel value that was automatically subtracted by the instrument. If automatic baseline correction was applied, the exported value can be represented as

$$Y_{exported,i}(d_c) = Y_{raw,i}(d_c) - B_i \quad (2.12)$$

Negative exported values in this setting are treated as evidence that the file may have been pre-subtracted by the instrument. Files containing systematic negative intensities are therefore flagged as likely affected by automatic baseline correction. For these flagged files, the preprocessing pipeline reverses the subtraction by adding the designated baseline channel back to the affected intensity channels on a row-wise basis:

$$Y_{reversed,i}(d_c) = Y_{exported,i}(d_c) + B_i \quad (2.13)$$

This reversal assumes that the negative values arose from automatic baseline subtraction rather than from detector malfunction, file corruption, or another measurement artifact. After reversal, each file is rechecked for remaining negative intensities. Files that still contain negative values after this correction are excluded from distributional modeling.

2.5.3 Timestamp Reset Correction

Some BCA files exhibit discontinuities in the recorded time sequence due to instrument logging resets or buffer flush events. In a valid continuous measurement segment, the timestamp variable, `time_sec`, should be monotonically nondecreasing within each file. A timestamp reset is identified when the current timestamp is smaller than the preceding timestamp:

$$t_i < t_{i-1} \quad (2.14)$$

where t_i denotes the value of `time_sec` for observation i .

For each BCA file, the full timestamp vector is scanned for monotonicity violations of the form shown in Equation (2.16). Each violation is treated as a reset boundary. If reset boundaries occur at indices

$$r_1, r_2, \dots, r_k \quad (2.15)$$

then the retained portion of the file is the final monotonic segment:

$$\{t_{r_k}, t_{r_k+1}, \dots, t_n\} \quad (2.16)$$

Equivalently, all observations before the final reset boundary are discarded, and only the observations from the final continuous timestamp segment are retained for analysis.

This rule preserves the most recent uninterrupted measurement window within each affected BCA file. The final segment is retained because it is assumed to represent the stabilized portion of the instrument run after any logging interruption or restart event. This correction ensures that downstream analyses are based on temporally coherent BCA observations within each retained file segment.

2.5.4 Cleaned Data Output

The preprocessing pipeline produces cleaned data outputs that support both pooled statistical modeling and file-level diagnostic review. The primary output is the long-format dataset, `bca_long.csv`. This file is the normalized analysis table in which each row represents one valid BCA observation at a specific timestamp and dwell time.

The preprocessing pipeline also saves wide-format outputs, `bca_wide.csv` and `dataframes.pkl`. These files preserve the original per-file, multi-channel structure of each BCA run. In the wide format, each row corresponds to a timestamp within a BCA file, and the intensity columns correspond to the six Prisma Pro dwell-time channels. This structure is useful for file-level quality checks, timestamp diagnostics, comparison across parallel dwell-time streams, and verification of preprocessing decisions.

Together, the long-format and wide-format outputs provide complementary views of the cleaned BCA data. The long-format file is used for pooled distributional analysis, dwell-time comparisons, model fitting, and summary statistics. The wide-format files are retained for diagnostics that require the original channel layout within each BCA run. These cleaned outputs are used in the downstream analysis notebooks, and their full data structures are documented in the data dictionary in Appendix C.

2.6 Aims of the Report

1. To document the data-processing rules that produce a clean BCA dataset suitable for distributional analysis.
 2. To describe the statistical procedures used to verify which features of the noise distribution are empirically supported and which are not, before any parametric or non-parametric model is chosen.
 3. To report quantitative findings on marginal distributions, dwell-time dependence, and any identified anomalies.
 4. To define and fit candidate noise models and to report their parameters and goodness of fit.
 5. To compare the models on a common set of validation criteria and to recommend a single method for further exploration in its application to downstream baseline correction.
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3. Preliminary Analysis

3.1 Numerical Summaries

The cleaned BCA dataset contains baseline-current measurements recorded at the six Prisma Pro dwell times defined in Equation (2.6). Each dwell-time group contains

$$n_d = 82,000 \tag{3.1}$$

observations recorded at each dwell time d in the Prisma Pro dwell-time grid. This produces a balanced dataset for comparing the empirical behavior of baseline noise across acquisition settings. Because each dwell-time group has the same sample size, differences in the observed summary statistics are not driven by unequal group sizes.

Table 1 reports the median, mean, and standard deviation of the measured baseline current at each dwell time. The summaries show a clear dwell-time dependence. As dwell time increases, the median, mean, and standard deviation all decrease. For example, the median current decreases from 2.986×10^{-14} A at $d = 0.128$ seconds to 5.256×10^{-15} A at 4.096 seconds. Similarly, the mean decreases from 3.928×10^{-14} A to 5.642×10^{-15} A, while the standard deviation decreases from 3.487×10^{-14} A to 3.703×10^{-15} A over the same dwell-time range.

Dwell Time d seconds	n	Median Amperes	Mean Amperes	Standard Deviation Amperes
0.128	82,000	2.986×10^{-14}	3.928×10^{-14}	3.487×10^{-14}
0.256	82,000	1.956×10^{-14}	2.490×10^{-14}	2.139×10^{-14}
0.512	82,000	1.266×10^{-14}	1.565×10^{-14}	1.292×10^{-14}
1.024	82,000	8.607×10^{-15}	1.025×10^{-14}	7.973×10^{-15}
2.048	82,000	6.363×10^{-15}	7.162×10^{-15}	5.128×10^{-15}
4.096	82,000	5.256×10^{-15}	5.642×10^{-15}	3.703×10^{-15}

Table 1. Summary statistics for intensity measurements by dwell time

This pattern indicates that longer dwell times are associated with lower observed baseline-current levels and reduced variability. Dwell time should therefore be treated as an important explanatory feature in later distributional modeling rather than as a secondary acquisition detail. In particular, the systematic decrease in both central tendency and spread suggests that a single pooled noise distribution would not adequately represent the baseline-noise process across all dwell times.

At this stage, the median and interquartile range are more reliable summaries than the mean and standard deviation because the raw current distribution is right-skewed and sensitive to upper-tail observations. The mean and standard deviation remain useful descriptive statistics, but model assessment should rely heavily on distributional diagnostics that are robust to skewness and tail behavior.

3.2 Exploratory Visualizations

On the raw scale, the distributions are concentrated close to zero with long upper tails. The raw scale histograms show an immediate intense spike and tails that might be better represented after transformation onto the $\log_{10}$ scale.

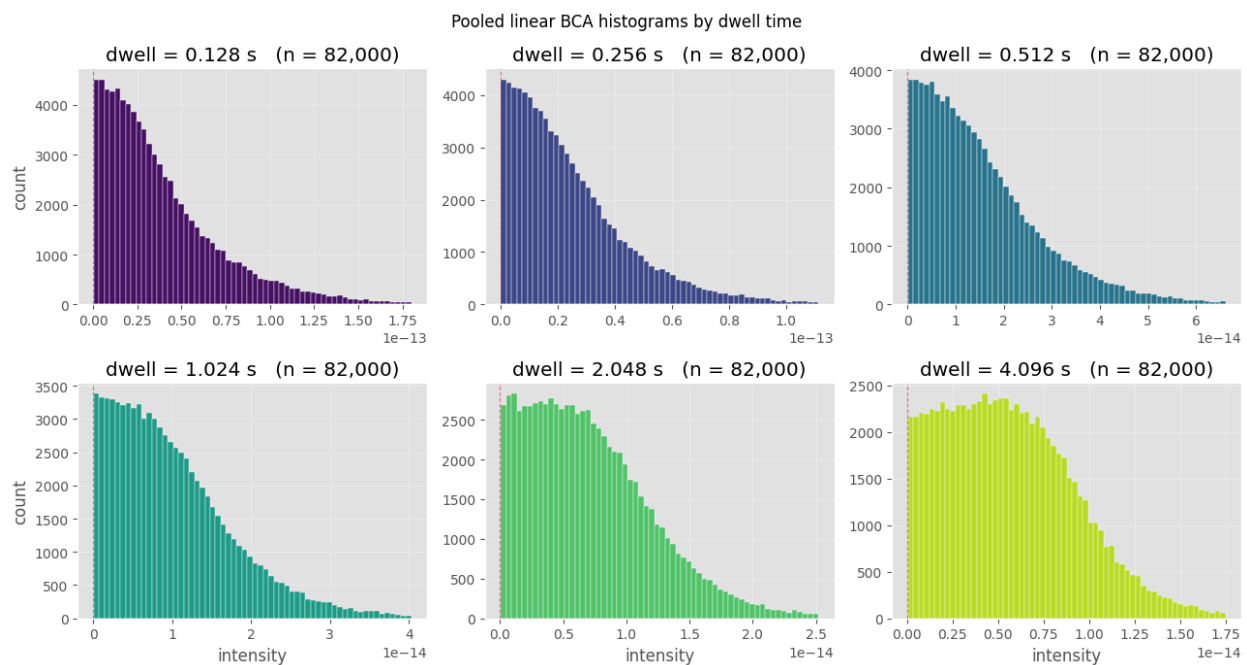


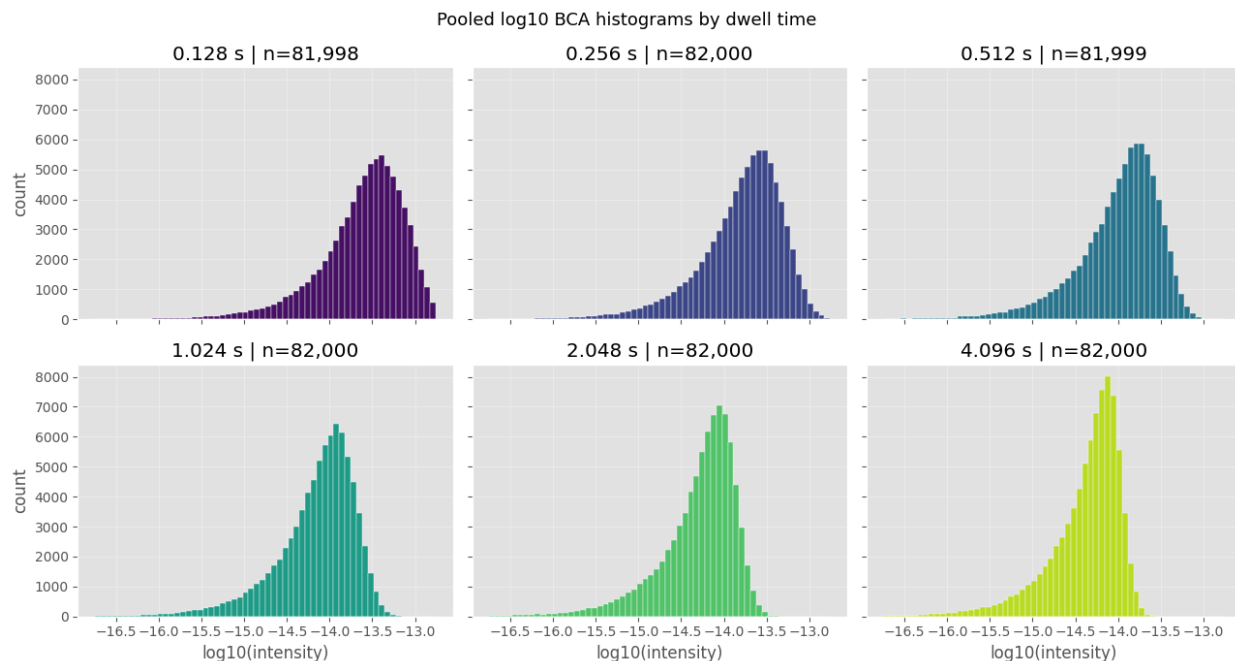
Figure 1. BCA noise histograms by dwell time on linear scale

The raw-scale data are highly compressed near zero and have long right tails. This supports using a positive-domain model and motivates examining the data on a logarithmic scale.

Three observations are excluded from the $\log_{10}$ transformed analysis because they were measured to have an intensity of 0 amperes. The $\log_{10}$ transform requires $Y > 0$ because $\log_{10}(0)$ is undefined. The three removed observations were:

Lab	Run ID	Dwell Time	Time Sec
glasgow	He 671	0.128	7739.996

salzburg	He 12352	0.512	33324.639
salzburg	He 12810	0.128	6549.218

Table 2. Removed observations before $\log_{10}$ transformation.Figure 2. BCA noise histograms by dwell time on $\log_{10}$ scale

Transforming the data onto the $\log_{10}$ scale makes the distributions easier to compare across dwell times. The centers shift steadily left as dwell time increases, and the spreads narrow, indicating that both location and scale depend on dwell time. However, the $\log_{10}$ histograms and Q-Q diagnostics still show skewness and tail departures, so the transformed data should not be treated as normally distributed.

3.3 Verification of Assumptions

3.3.1 Positive Domain and BCA Purity Assumption

The parser identifies BCA files from their repeated mass labels, then applies the automatic-subtraction reversal when negative values indicate that the acquisition software pre-subtracted a channel. After this step, no negative BCA intensities should remain in the long table.

After automatic-subtraction reversal, files were rechecked for remaining negative intensities. Any file with persistent negative values after reversal was excluded from distributional modeling or flagged for manual review. This preserves the positive-domain assumption required for logarithmic modeling.

The cleaned long table contains no negative BCA intensities, but three zero-valued observations remain. These observations are valid raw-scale records but are excluded from log-scale analyses because $\log_{10}(0)$ is undefined.

3.3.2 Log-Normality Assumption

Dwell	Mean $\log_{10}$ intensity	SD $\log_{10}$ intensity
0.128	-13.612	0.506
0.256	-13.802	0.499
0.512	-13.995	0.491
1.024	-14.167	0.479
2.048	-14.306	0.460
4.096	-14.394	0.446

Table 3. Summary statistics for positive $\log_{10}$ intensity measurements by dwell time

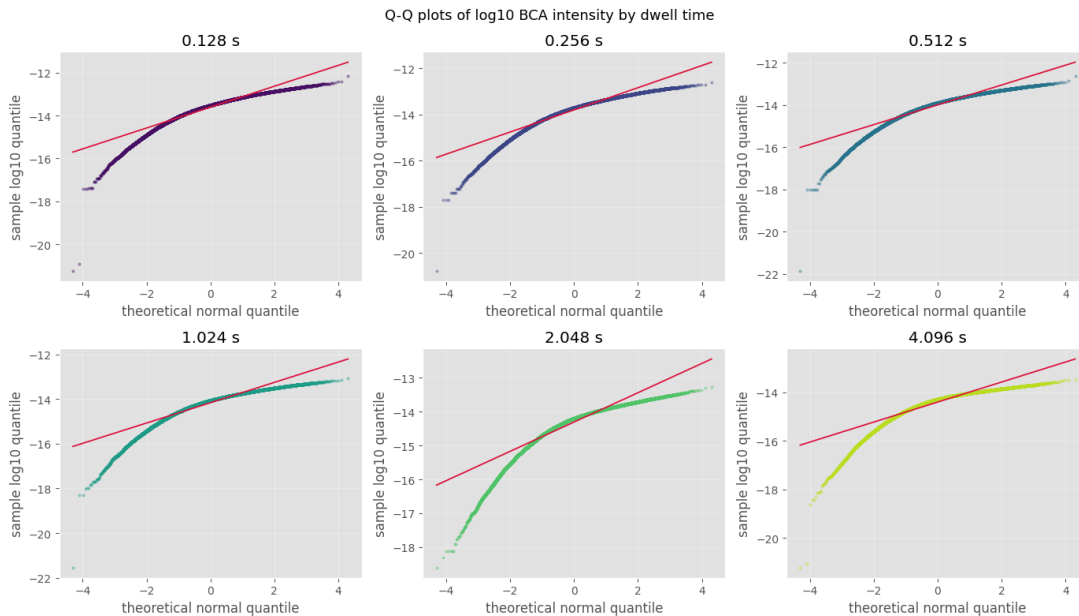


Figure 3. Q-Q plots of BCA noise by dwell time on $\log_{10}$ scale

The Q-Q plots show systematic departures in the tails. Therefore, the $\log_{10}$ transformed data do not align strongly enough with a normal distribution to justify a log-normal model as the final distributional assumption. The tail departures, together with the remaining asymmetry in the $\log_{10}$ histograms indicate that the transformed data remain skewed and heavy-tailed.

A simple log-normal model is still useful as a diagnostic reference because the $\log_{10}$ transformation makes the distribution more interpretable than the raw ampere scale. However, the remaining Q-Q departures motivate a more flexible log-scale distribution, leading to the proposed log-skew-t model.

3.3.3 Identical-Distribution Assumption Across Dwell Times

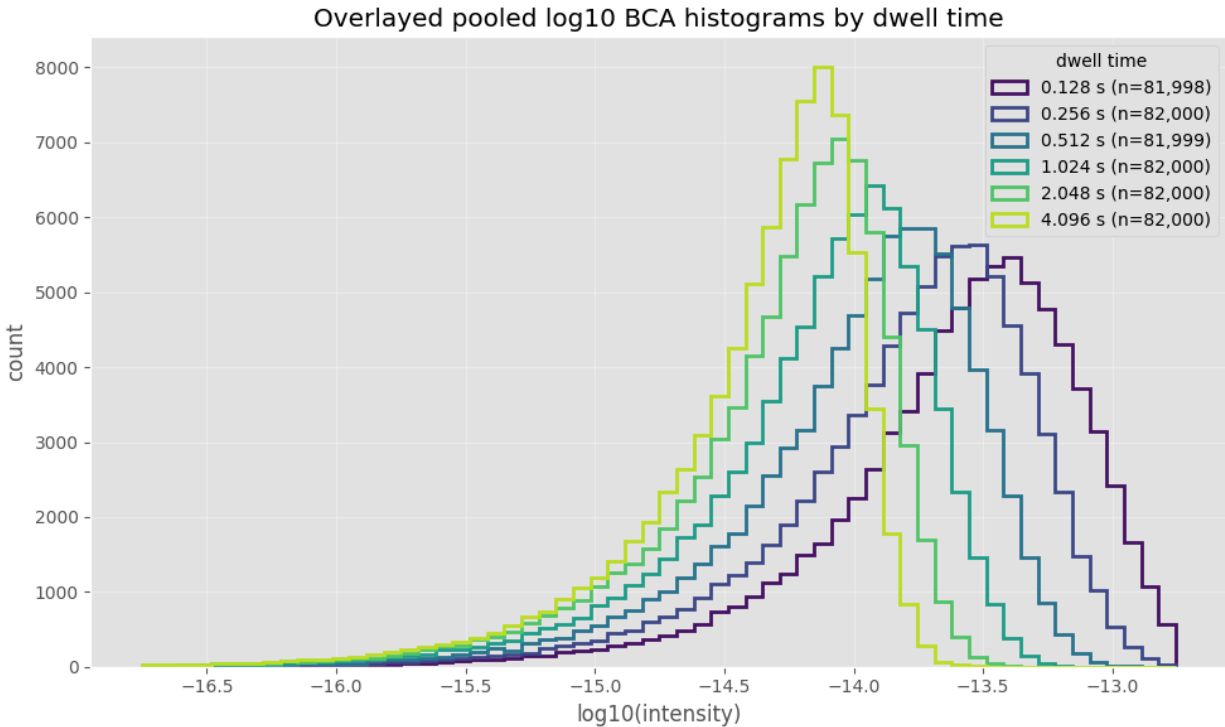


Figure 4. Stacked histograms of BCA noise by dwell time on $\log_{10}$ scale

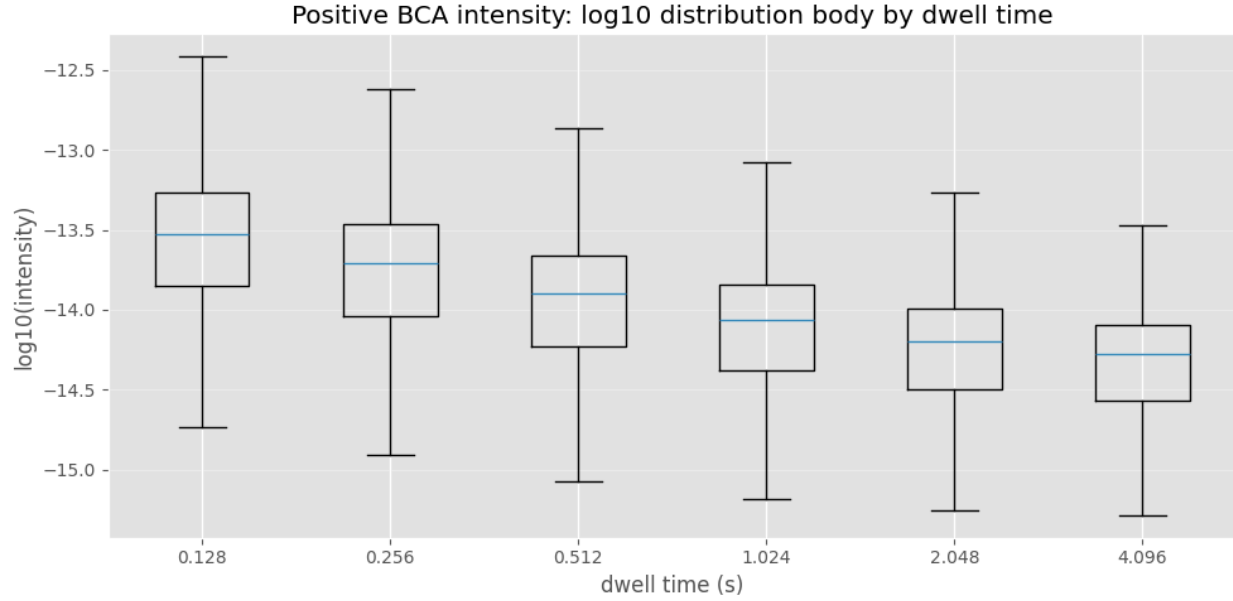


Figure 5. Box-and-whisker plots of BCA intensity on $\log_{10}$ scale by dwell time

The identical-distribution assumption is not supported. Center, spread, and tail summaries all change by different amounts across dwell times, therefore a single pooled distribution would obscure meaningful dwell-time structure.

4. Methodology

A channel, c_i , is one mass-spectrometer measurement stream, with an observation i being one recorded current value from one channel and one cycle. Let Y_i denote a single BCA intensity observation, d_i its dwell time, and t_i or time_sec the elapsed time stamp during the recorded measurement.

In this section, $\log(\cdot)$ denotes the natural logarithm. The $\log_{10}$ scale used in the visualizations and diagnostics were used for readability and interpretation. The model was actually fit on the natural log of intensity. Therefore, fitted model parameters refer to the natural-log scale, whereas the plotted distribution comparisons and cumulative probability diagnostics refer to the $\log_{10}$ scale.

4.1 Reparameterizing Log-Skew-t Distribution

Preliminary analysis showed that the distribution of baseline noise varies systematically with dwell time. As dwell time increases, both the center and spread of the baseline-current distribution decrease, and the tail behavior also changes. A single pooled distribution is therefore not appropriate for all dwell times.

To account for this dependence, the original Azzalini log-skew-t model described in Appendix D was reparameterized so that location, scale, and tail-heaviness vary smoothly with dwell time. The skewness parameter is estimated as one shared shape parameter across all dwell times. This structure captures the observed dwell-time dependence while avoiding a fully unconstrained fit for every dwell-time group.

Previous attempts to fit an unconstrained log-skew-t distribution suggested possible degeneracy between skewness and tail-heaviness parameters. Those concerns are discussed further in Appendix E. In the present model, dwell-time dependence is assigned to the location, scale, and tail-heaviness functions, reducing the burden on the shared skewness parameter to explain systematic changes across dwell times.

Let Y_i denote a positive baseline noise measurement and let d_i denote the associated dwell time. The model assumes

$$\log(Y_i) \sim \text{Skew-}t(\xi\text{BCA}(d_i), \omega\text{BCA}(d_i), \alpha, \nu\text{BCA}(d_i)) \quad (4.1)$$

where $\xi\text{BCA}(d_i)$ is a dwell-time dependent location function, $\omega\text{BCA}(d_i)$ is a dwell-time dependent scale function, α controls skewness, and $\nu\text{BCA}(d_i)$ is a dwell-time dependent tail-heaviness function.

The BCA data were used to estimate smooth dwell-time functions for location, scale, and tail-heaviness. These functions were estimated jointly by maximum likelihood rather than fixed from empirical summaries. The scale and degrees-of-freedom functions were transformed to keep the fitted parameters in the valid parameter space.

The fitted functions are parameterized as:

$$\xi\text{BCA}(d_i) = \beta_{\xi,0} + \beta_{\xi,1} \log(d_i) \quad (4.2)$$

$$\log(\omega\text{BCA}(d_i)) = \beta_{\omega,0} + \beta_{\omega,1} \log(d_i) \quad (4.3)$$

$$\log(\nu\text{BCA}(d_i) - 2) = \beta_{\nu,0} + \beta_{\nu,1} \log(d_i) \quad (4.4)$$

The log-scale residual for observation i is

$$z_i = \frac{\log(Y_i) - \xi\text{BCA}(d_i)}{\omega\text{BCA}(d_i)} \quad (4.5)$$

The log-likelihood contribution is

$$\ell_i = \log f_{ST}(z_i; \alpha, \nu\text{BCA}(d_i)) - \log(\omega\text{BCA}(d_i)) \quad (4.6)$$

where f_{ST} is the standard Azzalini-style skew-t density. The fitted parameters are the values that maximize the summed log-likelihood

$$\ell(\theta) = \sum_i \ell_i \quad (4.7)$$

where θ contains the coefficients defining $\xi\text{BCA}(d_i)$, $\omega\text{BCA}(d_i)$, $\nu\text{BCA}(d_i)$ and the shared skewness parameter α .

This approach allows the model to account for dwell-time-dependent changes in the center, spread, and tails of the baseline noise distribution while preserving flexibility for asymmetric and heavy-tailed behavior.

For each observation, the negative log-likelihood was computed from the standardized log-scale residual in Equation 4.5 and the Azzalini-style skew-t density in Equation 4.6. The scale and degrees-of-freedom functions were constrained to valid values by modeling $\log(\omega\text{BCA}(d_i))$ and $\log(\nu\text{BCA}(d_i) - 2)$, respectively. The optimizer minimized the summed negative log-likelihood over the coefficients for $\xi\text{BCA}(d_i)$, $\omega\text{BCA}(d_i)$, and $\nu\text{BCA}(d_i)$, and the shared skewness parameter α . Multiple starting values were used for the skewness and tail-heaviness parameters, and the final model was chosen as the converged fit with the smallest negative log-likelihood.

4.1.1 Estimating Dwell-Time Dependent Location Function Parameter

The dwell-time-dependent location function was estimated by maximum likelihood as part of the full log-skew-t model. It was not fixed to the empirical mean of log baseline intensity. In Equation 4.2 location is modeled as a smooth linear function of log dwell time.

The coefficients $\beta_{\xi,0}$ and $\beta_{\xi,1}$ were estimated jointly with the scale, skewness, and tail-heaviness parameters. This allows the fitted center of the *log* baseline-noise distribution to change continuously across dwell times.

4.1.2 Estimating Dwell-Time Dependent Scale Function Parameter

The scale function was also estimated by maximum likelihood. Since $\omega\text{BCA}(d_i)$ is a scale parameter, it must be strictly positive, so the model estimates $\log(\omega\text{BCA}(d_i))$ and then exponentiates back to the original scale. Equivalently,

$$\omega\text{BCA}(d_i) = \exp(\beta_{\omega,0} + \beta_{\omega,1}\log(d_i)) \quad (4.8)$$

Thus, $\omega\text{BCA}(d_i)$ is a fitted skew-t scale parameter, not the empirical standard deviation of $\log(Y)$ within each dwell-time group. Empirical log-scale summaries were used only as optimizer starting values and diagnostics.

4.1.3 Estimating Shared Skewness Parameter

Skewness was estimated as a single shared Azzalini-style skew-t shape parameter, α , across all dwell times. This parameter was optimized jointly with the dwell-time-dependent location, scale, and tail-heaviness functions.

Using one shared skewness parameter reduces instability in the fit. It also helps separate systematic dwell-time trends from general distributional asymmetry. Under this model, dwell time is allowed to affect center, spread, and tail-heaviness, while skewness is estimated from the full positive BCA dataset.

4.1.4 Estimating Dwell-Time Dependent Tail-Heaviness Function Parameter

Tail-heaviness was represented by the skew-t degrees-of-freedom parameter $\nu\text{BCA}(d_i)$. The tail-heaviness function was also estimated by maximum likelihood. Similarly, the skew-t degrees-of-freedom parameter must be positive, so this analysis constrained $\nu\text{BCA}(d_i) > 2$ so

that the fitted distribution has finite variance. This was done by modeling $\log(\text{vBCA}(d_i) - 2)$, which allows numerical optimization over real-valued coefficients while guaranteeing valid fitted values.

Equivalently,

$$\text{vBCA}(d_i) = 2 + \exp(\beta_{v,0} + \beta_{v,1} \log(d_i)) \quad (4.9)$$

The fitted values decrease as dwell time increases, from approximately 10.29 at 0.128 seconds to approximately 4.53 at 4.096 seconds. Because lower skew-t degrees of freedom correspond to heavier tails, this indicates heavier fitted tail behavior at longer dwell times.

4.2 Kernel Density Estimation

Kernel density estimation (KDE) was used as a nonparametric method for estimating the probability density function of the baseline-noise measurements. Unlike parametric models, KDE does not require the data to follow a specified distribution such as normal, lognormal, or skew-t. Instead, KDE estimates the distribution directly from the observed data by placing a smooth kernel function around each observation and summing those kernels to form an overall density curve.

Given observations $x_1, x_2, \dots, x_n$ the kernel density estimator is defined as

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right) \quad (4.10)$$

where $K(\cdot)$ is the kernel function, h is the bandwidth, and n is the number of observations.

The kernel function determines the shape of the small curve placed around each data point. A common choice is the Gaussian kernel, which places a bell-shaped curve centered at each observation. The bandwidth h controls the width of these curves. A smaller bandwidth produces a density estimate with sharper local variation, while a larger bandwidth produces a smoother estimate. In this way, KDE can be understood as a smoothed version of a histogram: instead of counting observations inside fixed bins, it spreads each observation's influence smoothly over nearby values.

Mathematically, each term

$$K\left(\frac{x-x_i}{h}\right) \quad (4.11)$$

measures how much observation x_i contributes to the estimated density at the point x .

Observations close to x contribute more, while observations farther away contribute less. The factor $\frac{1}{nh}$ normalizes the estimate so that the total area under the density curve is equal to 1.

Overall, KDE is used to provide an important check of whether the chosen parametric model adequately represents the observed baseline-noise behavior.

4.3 Application to the Baseline Correction Problem

The purpose of the reparameterized log-skew-t model is to estimate a baseline-noise distribution that matches the dwell time of each experimental signal channel. This is necessary because the preliminary analysis showed that baseline noise changes systematically with dwell time.

The live baseline channel remains useful, but it should not be treated as a universal cycle-level correction for every signal channel. In the experimental files considered here, the live baseline channel is measured at a fixed dwell time, while other signal channels may be measured at different dwell times. The BCA analysis shows that the center, spread, and tail behavior of baseline noise vary by dwell time.

For a signal channel with dwell time d_i , the BCA-calibrated baseline-noise distribution is

$$\log(N_i) \sim \text{Skew-}t(\xi\text{BCA}(d_i), \omega\text{BCA}(d_i), \alpha, \nu\text{BCA}(d_i)) \quad (4.12)$$

where N_i denotes the baseline-noise contribution for that channel. The functions $\xi\text{BCA}(d_i)$, $\omega\text{BCA}(d_i)$, and $\nu\text{BCA}(d_i)$ are estimated from the BCA data for each dwell time, while α is shared across dwell times.

This gives each channel a correction distribution that matches its dwell time while still pooling information across the full BCA dataset.

The fitted BCA distribution corresponding to each dwell time is used to estimate the expected baseline contribution for that channel. A simple first version of the correction can use a central summary of the BCA noise distribution at that dwell time, such as the median or mean. Because the BCA distributions are right-skewed and heavy-tailed, the median is often the more stable first-choice correction value.

The corrected signal for channel c_i can then be written as

$$Y_{true}(c_i) = Y_{raw}(c_i) - \widehat{N}(d_i) \quad (4.13)$$

where $Y_{raw}(c_i)$ is the recorded current for channel c_i , $\widehat{N}(d_i)$ is the estimated baseline-noise level for dwell time d_i , and $Y_{true}(c_i)$ is the baseline-corrected signal estimate.

When the median is used as the correction value,

$$\widehat{N}(d_i) = \text{median}[N_i | d_i] \quad (4.14)$$

where N_i follows the fitted BCA-calibrated log-skew-t distribution at dwell time d_i . The mean could also be used, but it is more sensitive to upper-tail behavior. For this reason, the median correction is recommended as the initial implementation. However, the final correction method should come from the dwell-time-specific BCA noise model, because that model directly accounts for the observed dependence of baseline noise on dwell time.

5. Full Analysis

5.1 Fitting the Baseline Noise Distribution

Overall, the BCA results show that baseline noise is strongly dwell-time dependent, right-skewed, and heavy-tailed. As dwell time increases, the center of the baseline distribution shifts downward and the spread generally narrows. This supports the reparameterized approach described in [Section 4.1](#), where the location, scale, and tail-heaviness parameters are allowed to vary by dwell time.

The fitted reparameterized log-skew-t model used 491,997 positive observations and produced $AIC = 1,351,186.11$ and $BIC = 1,351,263.86$. Since only one fitted parametric model is reported here, these values should be interpreted mainly as baseline model-selection criteria for future comparison. Lower AIC/BIC values from alternative models would indicate improved fit after accounting for model complexity. On the natural-log model scale, the fitted parameters show systematic dwell-time structure, with $\xi_{BCA}(d_i)$, $\omega_{BCA}(d_i)$, and $\nu_{BCA}(d_i)$ all decreasing as dwell time increases, indicating lower central noise, narrower spread, and stronger tail behavior at longer dwell times.

The fitted model estimates the dwell-time-specific calibration functions from the BCA data, while estimating one shared skewness parameter across all dwell times. This allows the model to preserve the observed dwell-time structure while still pooling information across the full BCA

dataset. The shared fitted parameter was $\alpha = 0.0592$. The small positive value of α suggests only mild overall skewness.

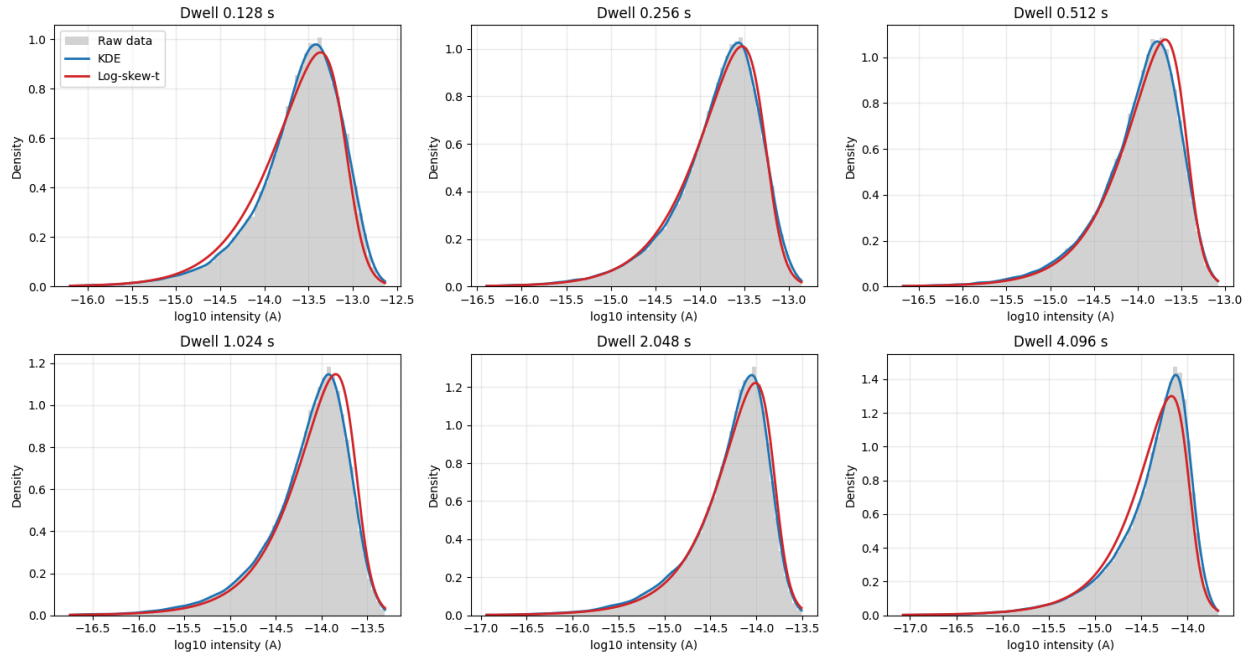


Figure 7. Comparison of dwell-time intensity distributions on the $\log_{10}$ scale

On the $\log_{10}$ intensity scale, the dwell-time distributions had broadly similar shapes but shifted left as dwell time increased, meaning longer averaging windows produced smaller baseline intensities. The KDE and fitted log-skew-t peaks were generally close, with the smallest mean absolute density gap at 0.256 seconds (0.013840) and the largest at 4.096 seconds (0.040103). This means the model is useful as a structured parametric approximation, but tail behavior and actual interpretability should continue to be checked carefully before the model is used for final baseline correction.

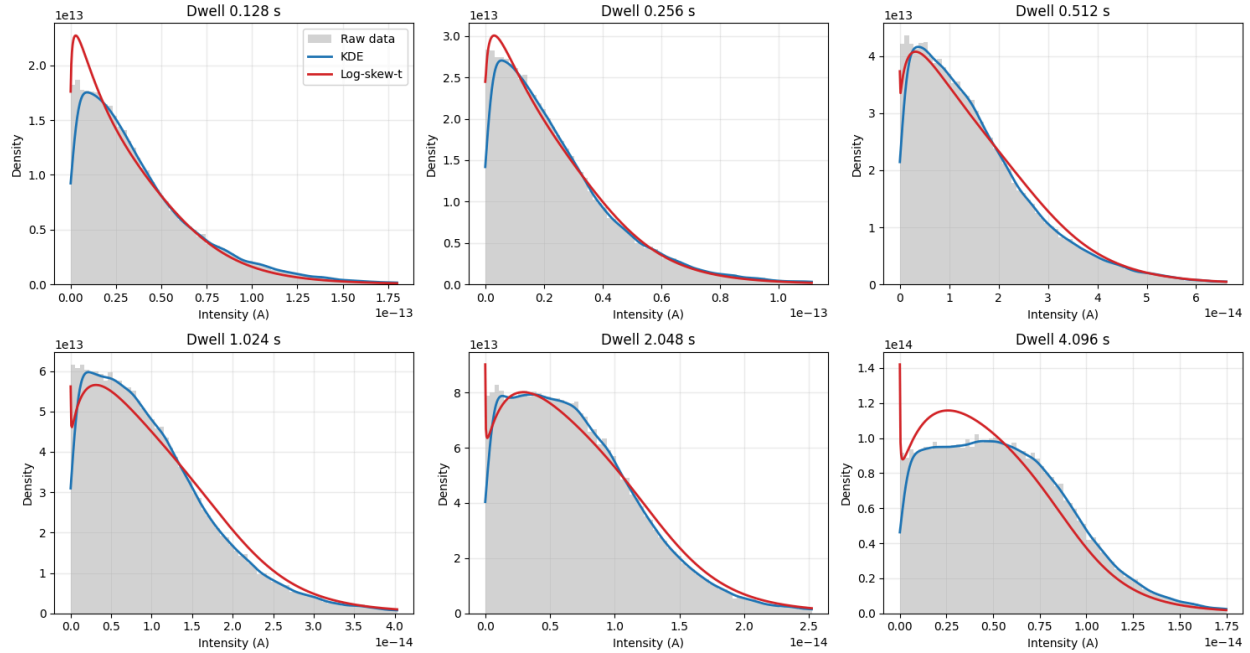
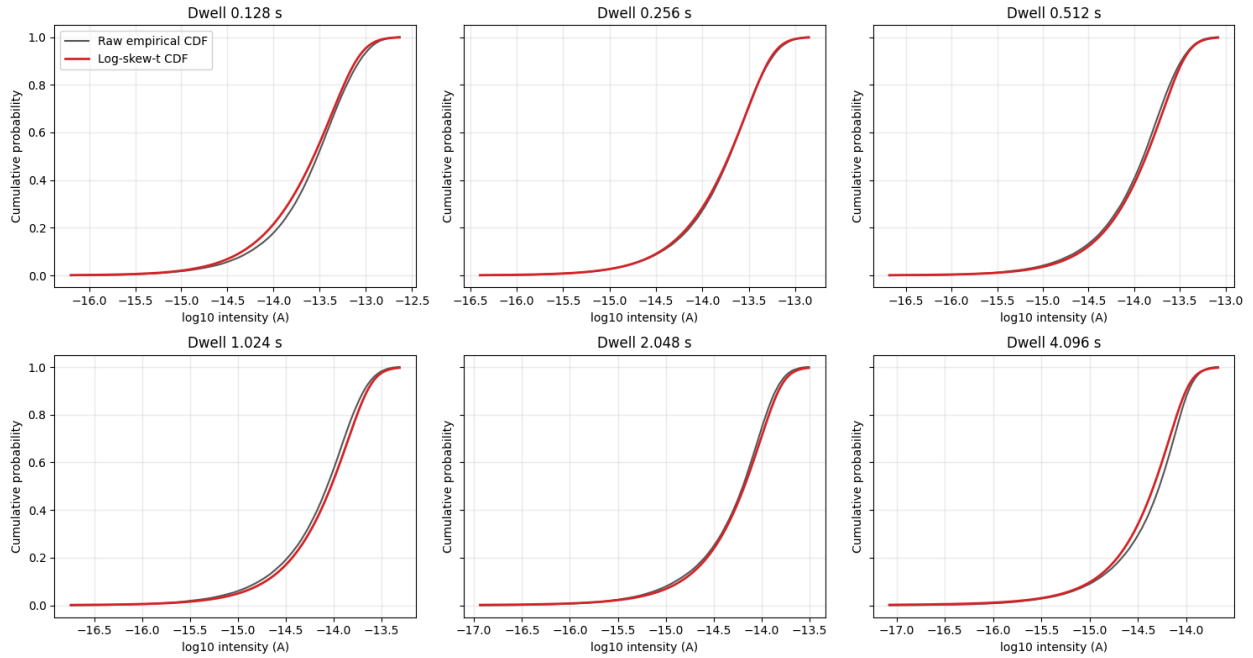


Figure 8. Comparison of dwell-time intensity distributions on the linear scale

The raw-scale plots are harder to compare visually because the distributions are concentrated near zero with long right tails, but they reinforce the same conclusion. The shortest dwell times have much wider and higher-intensity noise distributions, while longer dwell times compress the distribution toward lower currents.

Figure 9. Comparison of cumulative probability on $\log_{10}$ scale

The cumulative probability comparisons on the $\log_{10}$ scale shows how well the fitted log-skew-t model captures the full distribution, not just the density peak. The empirical CDF and fitted CDF are intended to track each other across the central probability range for each dwell time. Where the fitted curve accumulates probability too quickly or too slowly, the model is under or overestimating portions of the distribution, especially in the tails. Overall, the CDF view supports the use of a flexible positive-support model, but it also suggests that tail fit should remain a key diagnostic because baseline correction is sensitive to rare high-noise observations.

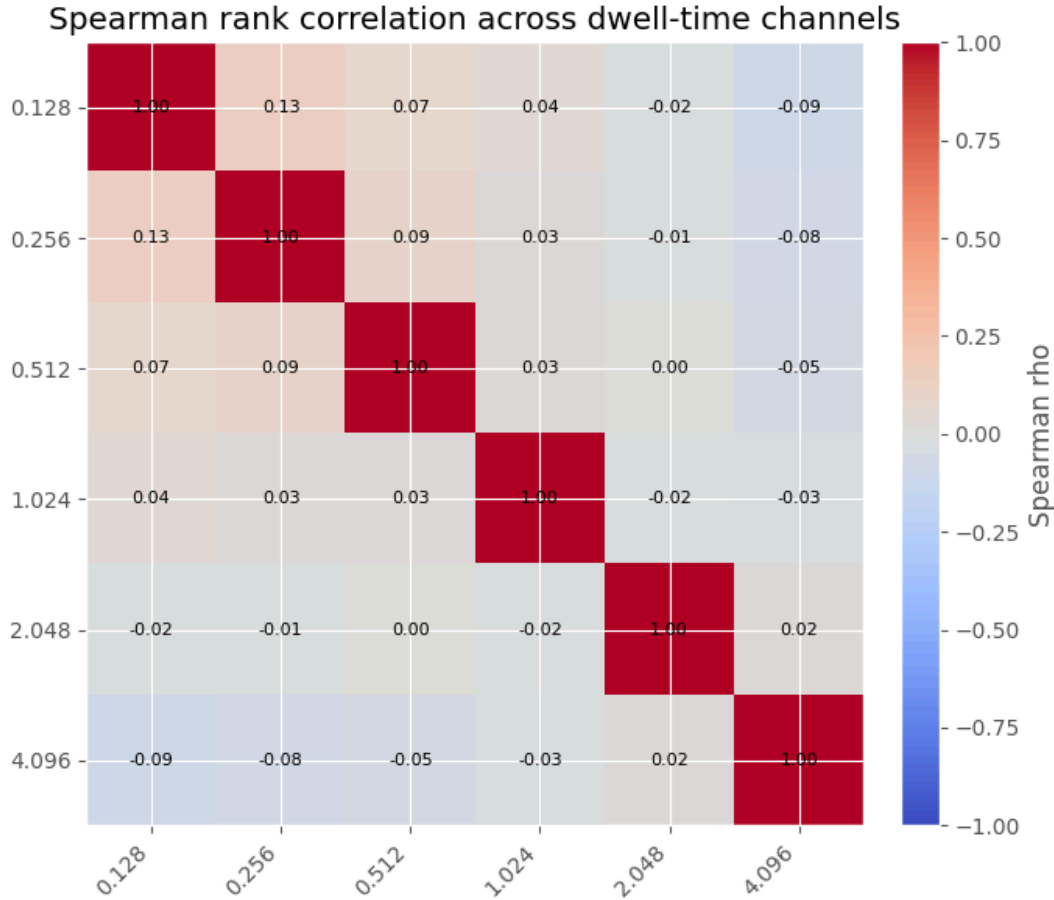


Figure 10. Spearman rank correlation across dwell-time channels

The Spearman rank correlation assesses the strength and direction of the monotonic relationship between two ranked or continuous variables. Cycle-aligned BCA files allow the six dwell-time channels to be compared within the same measurement cycle. The mean off-diagonal Spearman correlation was 0.007, with pairwise values ranging from -0.095 to 0.134. Therefore, the cross-dwell dependence is weak and uneven.

Overall, the analysis supports the main methodological conclusion: baseline correction should be dwell-time-specific. The BCA data provide enough information to estimate separate noise

behavior for each dwell time, and the reparameterized log-skew-t model provides a structured way to do this while still sharing information about skewness and tail behavior across the dataset.

6. Conclusions

6.1 Results Overview

This project shows that baseline noise in the mass spectrometer cannot be treated as a single constant background value. The amount and variability of noise depend strongly on dwell time, so subtracting one live baseline measurement from every signal channel risks applying the wrong correction.

By fitting a dwell-time-specific noise model, each experimental channel can eventually be corrected using the baseline distribution that matches its own dwell time. The reparameterized log-skew-t model is a useful framework for this because it captures the main dwell-time shift in the center, spread, and tail-heaviness of the data while still allowing for skewness.

The model should not be treated as final without caution. Diagnostic plots showed that the fitted log-skew-t distribution follows the main distributional shape but does not perfectly match the tails.

In summary, the BCA analysis supports a dwell-time-specific baseline correction strategy. This approach is more consistent with the observed behavior of the instrument and should reduce the risk of overcorrecting or producing physically impossible negative signals in future experimental analyses.

6.2 Measurement and Design Limitations

Analysis was limited to the available dwell-time settings: 0.128, 0.256, 0.512, 1.024, 2.048, and 4.096 seconds. As a result, any model describing the relationship between dwell time and baseline noise should be interpreted as an approximation based on only the observed settings.

Second, the baseline current distributions were highly skewed and contained upper-tail behavior. This makes the choice of distributional model important. While log-skew-t and kernel density approaches provide flexible ways to model the data, no single model can likely perfectly capture all features of the baseline-noise distribution.

Finally, the analysis focused on statistical characterization of baseline noise rather than direct correction of unknown samples. The results help describe how baseline noise behaves and how it changes with dwell time, but additional validation would be needed before applying the model as a routine correction method in production workflows.

6.3 Future Recommendations

Future work should collect baseline measurements across additional dwell-time settings to better understand whether the observed dwell-time relationship follows a smooth functional pattern.

Future work should also connect the baseline-noise model to practical correction workflows. Once the distribution is well characterized, the next step would be to evaluate how baseline-noise estimates affect downstream measurement uncertainty, signal correction, and final geochronologic interpretation.

7. References

Azzalini, A. and Capitanio, A. (2003), Distributions generated by perturbation of symmetry with emphasis on a multivariate skew t-distribution, *Journal of the Royal Statistical Society Series B*, 65, 367–389.

Higham, N. J. (1998), *Handbook of Writing for the Mathematical Sciences*, 2nd edn., Society for Industrial and Applied Mathematics, Philadelphia.

University of Manchester School of Mathematics, *Statistical Report Writing*

Oregon State University Department of Statistics, *Writing a Statistical Report for STAT 411/511*

8. Appendix

A. Index

Symbol	Meaning	Notes	Section
Y_{raw}	Recorded raw measurement after any instrument pre-correction has been reversed or disabled.	Contains true signal plus baseline noise.	§2.1 , §2.2 , §2.3 , §2.5.2 , §4.3 , §6

Y_{signal}	True ion signal of interest.	The target quantity after baseline correction.	§2.1 , §2.2 , §2.3 , §6
Y_{true}	Baseline-corrected true signal estimate.	Baseline noise subtracted from raw measurement to capture true signal.	§4.3 , §6
N or $N_i(d)$	Baseline-noise contribution.	Noise remains present even when no sample ions are present.	§2.1 , §2.2 , §2.3 , §4.3 , §6
Y_i	A single positive BCA baseline-noise observation.	Used in the log-skew-t model.	§4 , §4.1 , Appx. C
d , d_c , or d_i	Dwell time.	d_c indexes channel dwell time d_i indexes obs. dwell time	§2.1 , §2.2 , §2.3 , §3.1 , §4 , §4.1 , §4.3 , Appx. F
c or c_i	Measurement channel.	One mass-spectrometer measurement stream.	§2.3 , §4 , §4.3
t , t_i , or time_sec	Timestamp or elapsed time within a BCA run.	Used for timestamp reset correction and file parsing.	§2.4 , §2.5.1 , §2.5.3 , §2.5.4 , Appx. B
B_i	Baseline channel value automatically subtracted by the instrument.	The 5th column is used to reverse automatic baseline correction.	§2.5.2
n or n_d	Number of observations.	$n = 492000$ total clean obs. $n_d = 82000$ obs. per dwell time.	§2.4 , §3.1 , §4.2 , §5.1
$X = \log(Y)$	Log-transformed intensity.	$\log(\cdot)$ denotes the natural logarithm in the model.	§4 , Appx. C
$\log_{10}(Y)$	Base-10 log-transformed intensity.	Used for visualizations and diagnostics, not fitted model parameters.	§3.2 , §3.3.1 , §3.3.2 , §3.3.3 , §4 , §5.1

$\xi_{BCA}(d_i)$	BCA-calibrated natural-log-scale location function.	Estimated by maximum likelihood of BCA data.	§4.1 , §4.1.1 , §5.1 , Appx. C
$\omega_{BCA}(d_i)$	BCA-calibrated natural-log-scale scale function.	Estimated by maximum likelihood of BCA data.	§4.1 , §4.1.2 , §5.1 , Appx. C
$\nu_{BCA}(d_i)$	BCA-calibrated degrees-of-freedom function.	Estimated by maximum likelihood of BCA data. Lower values indicate heavier tails.	§4.1 , §4.1.4 , §5.1 , Appx. C
α	Shared skewness parameter in the skew-t model.	Estimated across all dwell times. Controls distributional asymmetry after log transform.	§4.1 , §4.1.3 , §4.3 , §5.1 , Appx. C , Appx. D
$z_i = \frac{\log(Y_i) - \xi_{BCA}(d_i)}{\omega_{BCA}(d_i)}$	Standardized residual on the natural-log model scale.	Used inside the skew-t likelihood.	§4.1 , Appx. A
$f_{ST} \sim \text{Skew-}t(\xi, \omega, \alpha, \nu)$	Standard Azzalini-style skew-t density.	Density used in the log-likelihood.	§4.1 , Appx. C
ℓ_i or ℓ_θ	Log-likelihood terms and parameter vector.	ℓ_i is one obs. contribution, ℓ_θ is the summed likelihood, θ contains all fitted parameters.	§4.1 , §5.1
$K(\cdot)$	Kernel function used in KDE.	Usually Gaussian. Places a smooth curve around each observation.	§4.2 , Appx. A
$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x-x_i}{h}\right)$	KDE notation.	Kernel function, bandwidth, evaluation point, observed data point, and estimated density.	§4.2
$Y_{raw} = Y_{signal} + N$	Raw signal decomposition	Defines each mass spectrometer measurement as signal plus baseline noise.	§2.1 , §2.3 , §6

If $Y_{\text{signal}} = 0$, then $Y_{\text{raw}} = N$	BCA purity condition	Explains why BCA files directly measure baseline noise.	§2.2 , §3.3.1
$Y_{\text{raw}} - N < 0$	Positive current requirement.	Indicates accidental automatic correction or a physically implausible corrected signal.	§2.1 , §6
$\log(Y_i) \sim \text{Skew-}t(\dots)$	Reparameterized BCA model. Natural-log model for positive BCA observations.	Allows location, scale, and tail-behavior to vary by dwell time while sharing skewness.	§4 , §4.1 , §4.3 , §5.1 , Appx. C
$E[N(d_i)] = \text{median}[N_i]$	Median baseline correction rule.	Recommended initial correction summary because the distribution is right-skewed and heavy-tailed.	§4.3
AIC or BIC	Model-selection criteria.	Lower values indicate better relative fit among comparable models after complexity penalties	§2.6 , §5.1 , Appx. C
KDE	Kernel density estimation.	Nonparametric benchmark for distribution shape.	§4.2 , §5.1
<i>Spearman rank correlation</i>	Rank-based monotonic association measure.	Used to evaluate cross-dwell dependence between cycle-aligned channels.	§5.1
AMU	Atomic mass unit.	Mass setting used by the instrument.	§2.2 , §2.3 , §2.4 , Appx. B , Appx. G
$\text{summary} = a \times d_i^b$	Power-law dwell-time relationship.	a is scale factor, b is the exponent controlling dwell-time change.	Appx. F

B. Data Dictionary

Variable	Definition	Notes
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lab	Glasgow Salzburg Wuhan Nineamu	Used to help separate different labs due to concern of changes in intensity between them due to different practices when measuring
run_id	Name of dataset file	Main purpose in data processing and cleaning
dwel_time	Length of time intensity of a given amu is measured	A key variable when looking at intensity, Baseline noise is strongly dwell-time dependent.
amu	Atomic mass unit	This is was the filter is tuned for, commonly set to 5.5amu to test for baseline noise
time_sec	Length of time measurement of test	Used during Timestamp Reset Correction
intensity	Recorded in amperes (A), when a detector is struck by ion at a selected amu and produces an electrical current.	Main measurement, all values are baseline noise due to amu setting

C. Azzalini Log-Skew-t Distribution

The log-skew-t distribution was considered as a parametric model for baseline noise because the observed measurements may be positive-valued, right-skewed, and influenced by occasional extreme deviations. Traditional normal models assume symmetry and light tails, which may not adequately describe baseline noise if the data contains systematic skewness or outliers. The skew-t distribution extends the normal distribution by introducing parameters that control both asymmetry and tail heaviness. Applying the model on the logarithmic scale further allows the distribution to describe strictly positive measurements while stabilizing variance and reducing the influence of large observations.

Let Y represent a baseline noise measurement. Under a log-skew-t model, the transformed variable

$$f_{ST} = \log(Y) \quad (\text{C.1})$$

is assumed to follow a skew-t distribution:

$$f_{ST} \sim \text{Skew-}t(\xi, \omega, \alpha, \nu) \quad (\text{C.2})$$

where ξ is a location parameter, ω is a scale parameter, α controls skewness, and ν controls the heaviness of the tails.

Smaller values of ν allow for heavier tails, making the model more robust to extreme baseline values. When $\alpha = 0$, the distribution becomes symmetric, and when ν becomes large, the skew-t distribution approaches a skew-normal distribution.

This model is useful because it separates several important features of the baseline noise distribution. The location parameter describes the central tendency of the logged measurements, the scale parameter describes spread, the skewness parameter captures directional asymmetry, and the degrees-of-freedom parameter accounts for unusually large or small observations. This makes the log-skew-t model especially appropriate when the distribution is not well-described by a simple Gaussian assumption.

Model parameters may be estimated using maximum likelihood estimation. After fitting the distribution, goodness-of-fit is evaluated using visual diagnostics such as histograms with fitted density curves and Q-Q plots. Quantitative measures such as log-likelihood, AIC, or BIC may also be used to compare the log-skew-t model against alternative candidate distributions.

D. Adam's Previous Log-Skew-t Attempts

Adam's preliminary analyses showed that the log-skew-t distribution fit the BCA data well. However, his early fitting attempts suggested possible degeneracy among the shape parameters, particularly between the skewness parameter, α , and the tail-heaviness parameter, ν . In some fits, changes in these parameters appeared to compensate for one another, meaning the histogram could be fit well even when the individual parameter estimates were not uniquely meaningful.

Because the BCA measurements occur at extremely small current scales, approximately $\sim 10^{-16}$ to $\sim 10^{-13}$ amperes, the model-selection criterion was made intentionally strict. Likelihood-profile and bootstrap diagnostics did not indicate a broad $\frac{\alpha}{\nu}$ degeneracy in the reparameterized model.

Therefore, the full log-skew-t distribution was retained as the primary empirical model for baseline noise. The fitted skewness and tail-heaviness parameters are interpreted cautiously, and their reliability is evaluated using the identifiability diagnostics reported here.

E. Lab Specific Effects

Some labs have their mass spectrometer enclosed in strictly controlled and regulated environments, while others are not as carefully isolated from the external environment.

We can then pool over each dwell time to give a coarse lab-level audit to identify the differences, contributed by the lab location, in the baseline distribution.

Lab	n	Mean	Median	Standard Deviation
glasgow	84000	1.153×10^{-14}	8.644×10^{-15}	9.894×10^{-15}
nineamu	120000	2.131×10^{-14}	1.175×10^{-14}	2.640×10^{-14}
salzburg	198000	1.888×10^{-14}	1.003×10^{-14}	2.396×10^{-14}
wuhan	90000	1.303×10^{-14}	9.132×10^{-15}	1.303×10^{-14}

Table 4. Lab level summary statistic tables

These panels compare each lab across dwell time. Median checks center, IQR checks distribution body spread, and the 95th/99th percentiles check upper-tail behavior. Parallel curves suggest labs differ mostly by a stable offset. Crossings or one lab separating only in the tail suggest lab-specific distribution shape differences.

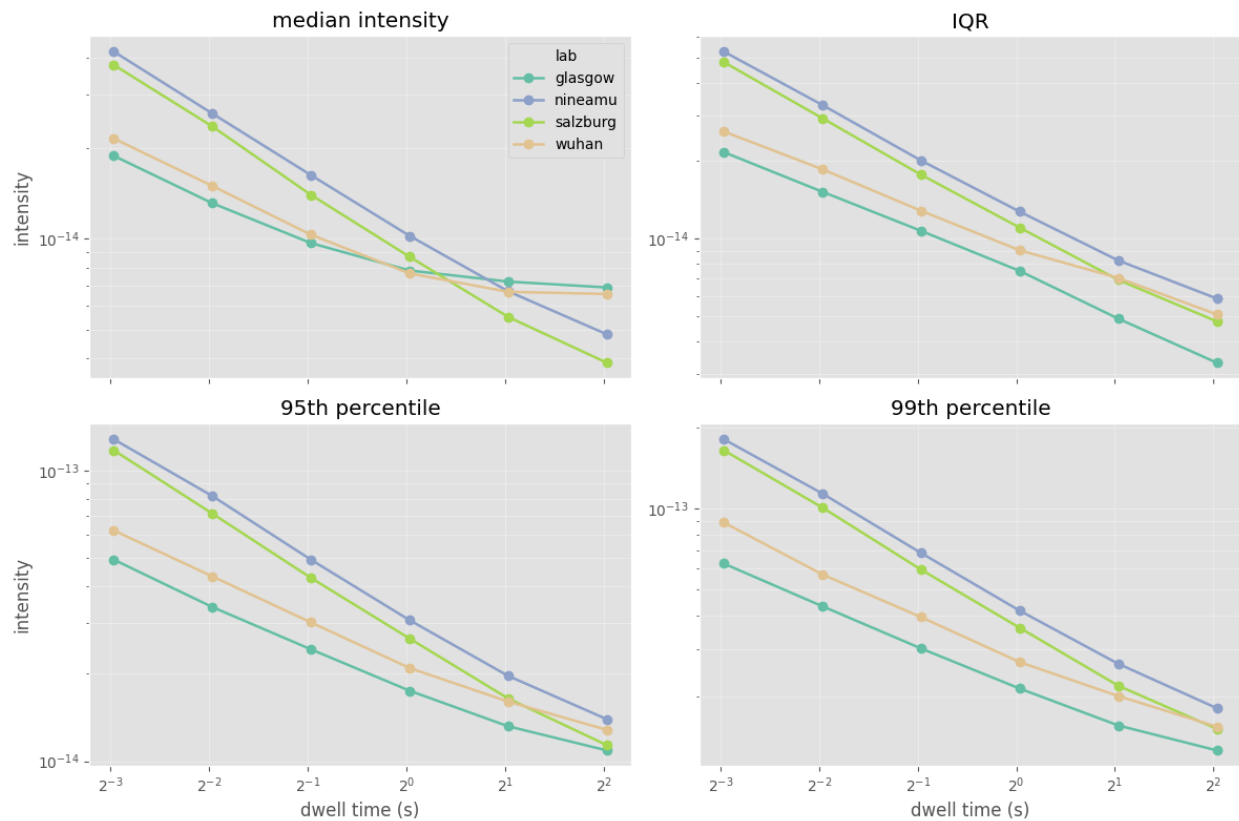


Figure 13. Lab-level center, spread, and tail curves

We can see there may be lab-specific differences in measured baselines noise, however there appears to be a noticeable linear relationship across all sites.

F. Dwell-Time Power Law Relationship

The dwell-time diagnostics test whether the positive BCA intensity summaries follow a power-law relationship of the form

$$summary = a \times d_i^b \quad (F.1)$$

where *summary* is the BCA intensity summary being modeled, such as the mean, median, standard deviation, IQR, 95th percentile, or 99th percentile. While *a* is the scale factor which sets the overall height of the curve. In this model, it is the predicted summary value when dwell time equals 1 second, because $1^b = 1$. Lastly, *b* is the power-law exponent. It controls how quickly the summary changes as dwell time increases.

This is useful because dwell time is expected to affect baseline noise systematically: if the relationship is close to a power law, then the distribution summaries should appear approximately linear on log-log axes, and the exponent *b* gives a compact description of how each summary changes as dwell time increases. The analysis focuses on positive intensities so the log transformation is well-defined and compares the mean, median, standard deviation, interquartile range, and upper-tail quantiles across the six dwell times.

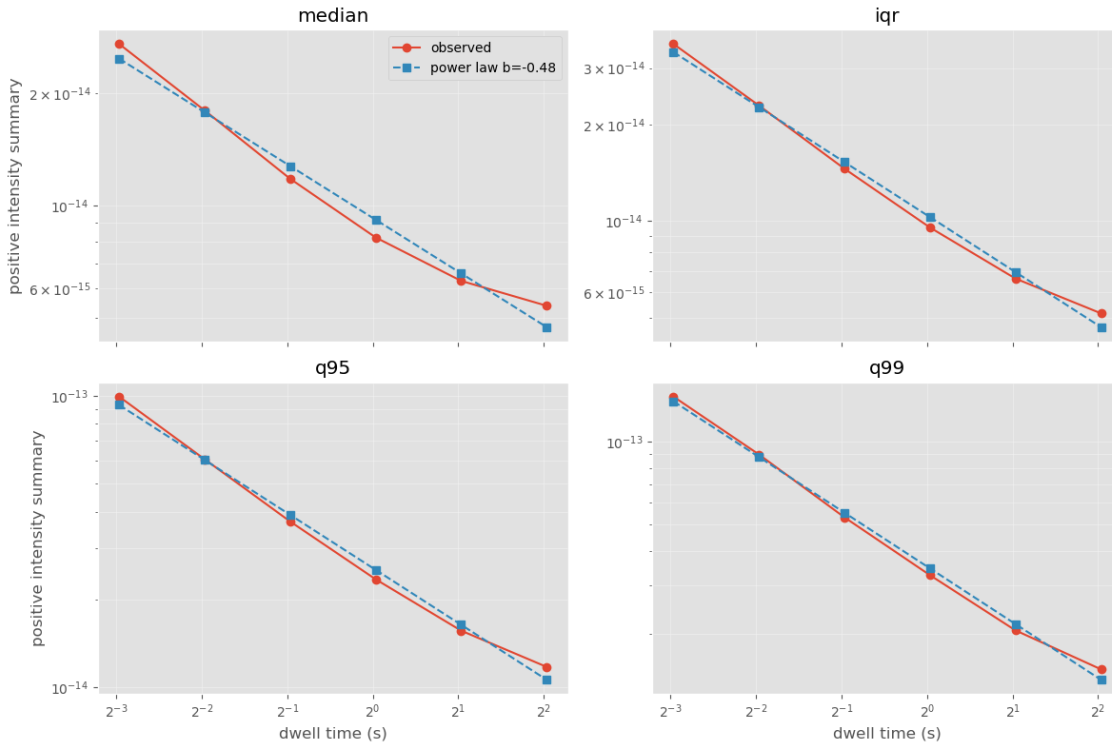


Figure 10. Log-log power-law fit diagnostic plots

The power-law fit diagnostic plots show that the main body and upper tail of the positive BCA distribution decline smoothly as dwell time increases. The fitted exponents are all negative, ranging from approximately -0.51 for the median to -0.67 for the 99th percentile, so longer dwell times are associated with lower positive baseline-noise summaries. The upper-tail summaries have more negative exponents than the median and IQR, which suggests that dwell time changes not only the location of the distribution but also its shape, with the upper tail shrinking somewhat faster than the central tendency.

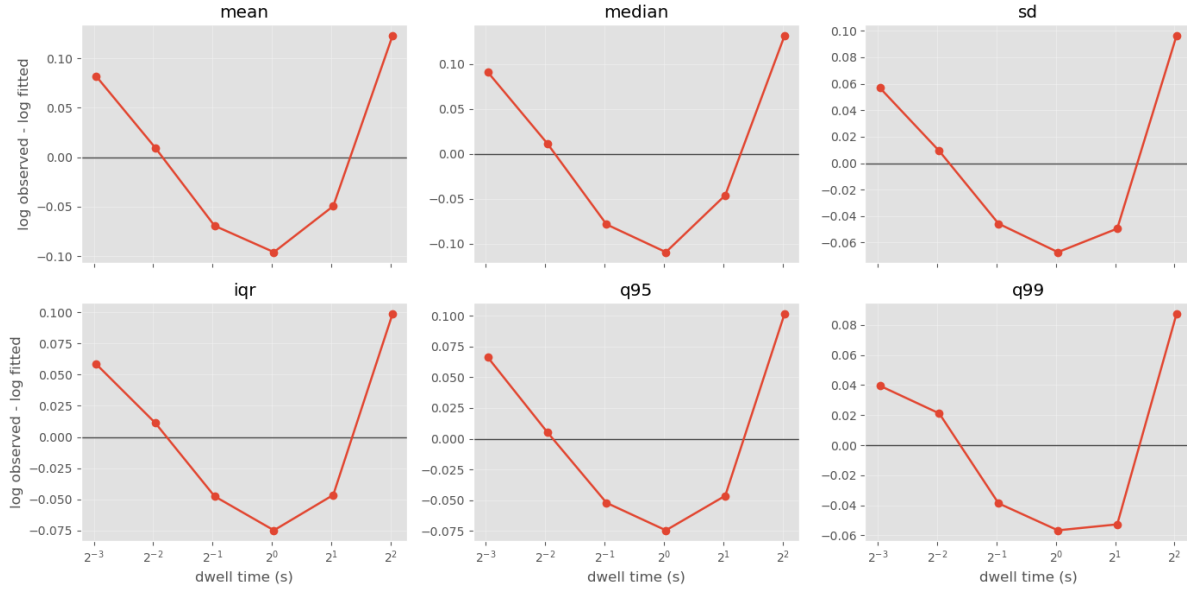


Figure 11. Power-law residuals plots

The residual plots show small but structured departures from a perfect power law. Across the summaries, residuals are generally positive at the shortest dwell time, negative through the middle dwell times, and positive again at the longest dwell time, with the largest positive residual usually at 4.096 seconds and the most negative residual around 1.024 seconds. This pattern means the power law is a useful first-order description of dwell-time dependence, but it does not fully remove dwell-time-specific structure. For downstream modeling, the residual pattern supports retaining dwell-time-aware empirical modeling rather than relying on a single pooled correction curve.

G. Investigating Differences in AMU Settings

An area of potential focus was potential changes that may occur with different amu's. As a result we received 20 different datasets from the same laboratory, with 10 of the dataset being set to the default amu of this paper of 5, and the other 10 being set to another amu that can serve the same purpose at 9 amu.

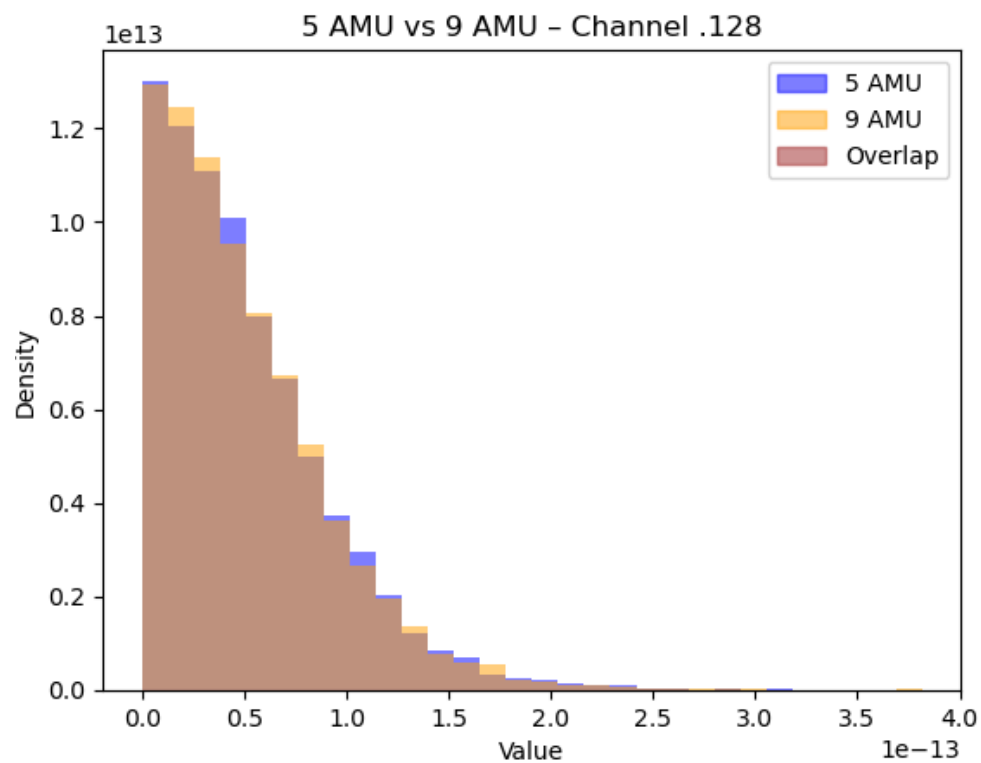


Figure 12: Overlapping Histogram 5vs9 amu (0.128 second dwell time)

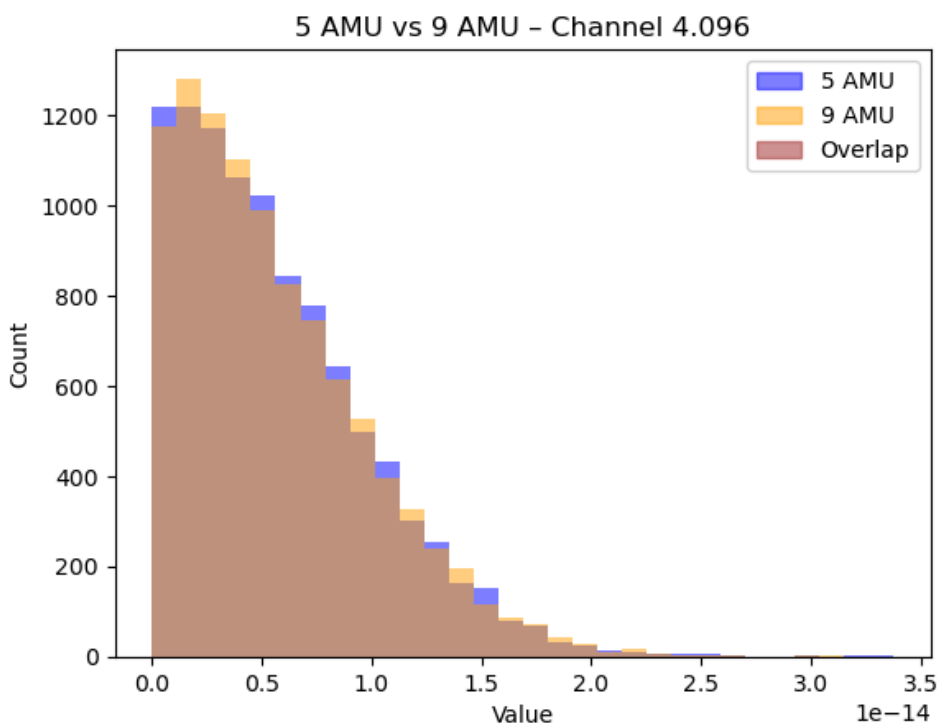


Figure 13: Overlapping Histogram 5vs9 amu (4.096 second dwell time)

Both plots show small changes that give little evidence that there are changes between the 2 amu's of 5 and 9. This casts doubt on the previous theory of changes of distribution of intensity between different amu's being measured. This in turn supports the reparameterized approach described in Section 4.3, where the location, scale, and tail-heaviness parameters are allowed to vary by dwell time. Plots at different dwell times paint a similar picture.